

Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Computational physicist with a PhD in Physics and ten years of experience building and running simulation codes for relativistic plasmas. Author of Paramo, an open-source Fortran library that solves the Fokker-Planck equation coupled to radiative transfer. Led an upgrade of the PatchworkMHD code on Frontera and used gradient-based optimization to calibrate model parameters against observations. Nine publications, three as first author, and Co-I on a funded NASA Fermi grant.

Skills

Simulation: Kinetic plasma modeling (Fokker-Planck), radiative processes (synchrotron, inverse Compton), radiative transfer, relativistic MHD, Lagrangian particle dynamics, numerical PDEs, finite difference/volume methods, numerical linear algebra, parameter estimation, uncertainty quantification

HPC & Parallel: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5

ML & Data: Scientific machine learning, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Languages: C/C++, Python, Fortran 95, Rust, R, Java, SQL, Julia, Shell

Tools: Git, Docker, Jenkins, QGIS, Paraview, VisIt

Experience

Independent Contractor

Jan 2026 — Present

Mercor

- Provide deep expertise in physics and numerical methods to improve LLM models for a top AI lab.

Spatial Data Scientist

May 2025 — Jan 2026

TealWaters

- Led optimization of scientific code for topographic terrain modeling, improving computational performance of elevation derivative calculations.
- Engineered Python tools for geospatial data processing and analysis; applied random forest ML for wetland probability mapping.

Software Engineer

Apr 2022 — Jan 2024

Paychex

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

Postdoctoral Research Associate

Feb 2021 — Apr 2022

Rochester Institute of Technology

- Led an NSF-sponsored upgrade of PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries, adding black hole spin without increasing runtime.
- Ran and benchmarked large production simulations on Frontera (TACC) and identified performance bottlenecks in the parallel code.
- Calibrated Paramo parameters against Fermi-LAT observations using gradient-based optimization, comparing simulation output with data and updating the model automatically.
- Collaborated on a multi-institutional HPC project on binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow

Oct 2018 – Nov 2020

Purdue University

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker-Planck equation for relativistic particle distributions coupled to radiative transfer. Parallelized with OpenMP and tested GPU offload with OpenACC.
- Implemented implicit PDE solvers, numerical integration, and support for particle distributions of arbitrary shape without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization.
- Co-I on two NASA Fermi Guest Investigator proposals, one funded (\$68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

Postdoctoral Research Fellow

Jan – Sep 2018

Universidad Michoacana de San Nicolás de Hidalgo

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

Graduate Research Assistant

Oct 2011 – Jul 2017

Universitat de València

- Independently developed Shell and Python pipelines to run the C-SPEV radiative transfer simulation code and process large HDF5 output datasets.
- Implemented numerical tools and data handling extensions to C-SPEV for computing both discrete and continuous spectra from particle distributions with arbitrary shape.
- Published 2 first-author papers on numerical simulations of relativistic plasma dynamics.

Projects

Blazar Population ML, github.com/altjerue/ilwikak-ml.

Jan 2026 — Present

Scientific ML (Python, scikit-learn, PyTorch)

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Forecasting blazar gamma-ray flares using a general-purpose time series foundation model, fine-tuned with Tleco-generated synthetic light curves.

WindsOfChange, github.com/UW-ForestMycobiomeLab/WindSporesHills.

Aug 2024 — Present

Open-Source Lagrangian Particle Simulation Library (R/C++)

- Developed C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Tleco, github.com/zkdavis/Tleco.

Jan 2024 — Sep 2024

Open-Source Physics Simulation Library

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Paramo, github.com/altjerue/paramo.

Oct 2018 — Apr 2022

Open-Source Physics Simulation Code (CONAHCYT Funded)

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.

Selected Publications (9 total, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, *ApJ* **976**, 182, (2024), arXiv:2405.17581.
- **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, *MNRAS* **501**, 4092–4102, (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, *MNRAS* **468**, 1169–1182, (2017), arXiv:1612.06383.

Education

Universitat de València

Valencia, Spain

Ph.D. in Physics. Excellent Cum Laude.

Jul. 2017

Universidad Michoacana de San Nicolás de Hidalgo

Morelia, Mexico

M.Sc. in Physics.

Sep. 2011

Universidad Autónoma del Estado de México

Toluca, Mexico

B.Sc. in Physics.

Dec. 2008